

Name:
Roll number:

Total Points: 15
Duration: 60 minutes

1. [CO1, CO2, CO3] Batch Normalisation and L2 Regularisation (0.5 + 0.5 points)

A network applies Batch Normalisation (BN) and L2 weight decay on all layers. It has been established that L2 exerts no regularising effect on BN-normalised layers. Which option correctly identifies the reason?

- (a) The L2 penalty gradient is projected onto a constant-norm weight manifold by BN, contributing zero net curvature to the regularised objective.
- (b) Per-batch renormalisation restores the pre-activation distribution after each weight update, neutralising the shrinkage induced by the L2 penalty.
- (c) For any $c > 0$, $\text{BN}(c\mathbf{W}; \mathbf{x}) = \text{BN}(\mathbf{W}; \mathbf{x})$, so L2 decay modifies $\|\mathbf{W}\|$ without altering the network output, affecting only the effective learning rate.
- (d) The affine parameters γ and β span the same functional space as $\mathbf{W}$, rendering the L2 constraint redundant with respect to the network's representational capacity.

Solution: Option (C) is correct: since $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}$ scale proportionally with $\|\mathbf{W}\|$,

$$\text{BN}(\mathbf{W}\mathbf{x}) = \gamma \cdot \frac{\mathbf{W}\mathbf{x} - \mu_{\mathcal{B}}}{\sigma_{\mathcal{B}}} + \beta,$$

the substitution $\mathbf{W} \rightarrow c\mathbf{W}$ leaves the output identically unchanged for all $c > 0$. L2 decay thus operates on a degree of freedom to which the loss is insensitive, and its sole observable effect is an implicit modulation of the effective learning rate, $\eta_{\text{eff}} \propto \eta \|\mathbf{W}\|^{-2}$.

2. [CO1, CO2, CO4] Slack Variables and Misclassification (0.5 + 0.5 points)

In the soft-margin SVM, the slack variable for the i -th sample is $\xi_i = \max(0, 1 - y_i(\mathbf{x}_i^\top \boldsymbol{\theta} + b))$. Four regions relative to the decision boundary are identified as R1 ($\xi_i < 0$), R2 ($\xi_i = 0$, on L_1), R3 ($0 < \xi_i < 1$), and R4 ($\xi_i > 1$). Which classification status corresponds to R3?

- (a) Correct classification but with a margin violation: the sample lies between L_0 and L_1 , satisfying $y_i(\mathbf{x}_i^\top \boldsymbol{\theta} + b) > 0$ but < 1 , and contributing $\xi_i \in (0, 1)$ to the hinge loss.
- (b) Correct classification with the sample lying beyond the margin boundary, contributing zero to the hinge loss.
- (c) Misclassification: the sample lies on the wrong side of the decision boundary L_0 .
- (d) The sample lies exactly on the decision boundary L_0 , giving $\xi_i = 1$ and representing a borderline misclassification.

Solution: Option (A) is correct: R3 corresponds to $0 < \xi_i < 1$, meaning $0 < y_i(\mathbf{x}_i^\top \boldsymbol{\theta} + b) < 1$. The sample is correctly classified (on the right side of L_0) but violates the margin (lies between L_0 and L_1). This is a margin violation without misclassification, and the sample contributes positively to the hinge loss without being a true error.

3. [CO1, CO2, CO3] Occam's Razor Hypothesis (0.5+0.5 points)

Occam's Razor, when applied to machine learning, favors simpler models over complex ones. Which of the following mathematical formulations most directly encodes this principle?

- A. Selecting the model that achieves the highest training accuracy
- B. Adding a penalty term $\lambda \|\boldsymbol{\theta}\|$ to the loss function, where increasing λ shrinks the magnitude of learned weights
- C. Increasing the number of hidden layers to reduce bias
- D. Decaying the learning rate η over training iterations

Solution: Option (B) is correct: Regularization directly encodes Occam's Razor: the penalty term $\lambda \|\boldsymbol{\theta}\|$ discourages large weights, effectively preferring simpler (smaller-magnitude) parameter configurations.

4. [CO1, CO2, CO3] Evaluation Metrics

(3 points)

A multi-class classifier is evaluated on a test set of animal images across three classes: Cat, Dog, and Bird. The confusion matrix is:

True \ Pred	Pred		
	Cat	Dog	Bird
Cat	40	5	5
Dog	3	32	5
Bird	2	8	10

- (a) Compute the Precision, Recall, and F1-score for each class. Treat each class in turn as the positive class.
- (b) The overall accuracy of this classifier is 75.5%. A developer argues this is sufficiently high for deployment. Using your results from part (a), evaluate whether this argument is justified.

Solution:

(a) From the confusion matrix, the column sums give predicted positives and row sums give actual positives:

Cat:

- $TP = 40$, $FP = 3 + 2 = 5$, $FN = 5 + 5 = 10$
- $Precision = \frac{40}{40+5} = \frac{40}{45} \approx 0.889$
- $Recall = \frac{40}{40+10} = \frac{40}{50} = 0.8$
- $F1 = \frac{2 \times 0.889 \times 0.8}{0.889 + 0.8} = \frac{1.422}{1.689} \approx 0.842$

Dog:

- $TP = 32$, $FP = 5 + 8 = 13$, $FN = 3 + 5 = 8$
- $Precision = \frac{32}{32+13} = \frac{32}{45} \approx 0.711$
- $Recall = \frac{32}{32+8} = \frac{32}{40} = 0.8$
- $F1 = \frac{2 \times 0.711 \times 0.8}{0.711 + 0.8} = \frac{1.138}{1.511} \approx 0.753$

Bird:

- $TP = 10$, $FP = 5 + 5 = 10$, $FN = 2 + 8 = 10$
- $Precision = \frac{10}{10+10} = \frac{10}{20} = 0.5$
- $Recall = \frac{10}{10+10} = \frac{10}{20} = 0.5$
- $F1 = \frac{2 \times 0.5 \times 0.5}{0.5 + 0.5} = 0.5$

(b) The developer's argument is not justified. While the overall accuracy of 75.5% appears acceptable, the per-class metrics reveal a severely imbalanced performance. The Bird class has a Precision, Recall, and F1-score of only 0.5, meaning the classifier is completely random for bird class while the Cat class dominates the dataset (50 samples) and is classified well, inflating the overall metric.

5. [CO1, CO2, CO3] Dropout, Batch-Normalization

(3 points)

Dropout randomly sets each neuron's output to 0 with probability p during training.

- (a) A network has $n = 4$ neurons in a hidden layer. With $p = 0.5$, how many distinct sub-networks can be sampled? Argue that training with dropout is approximately equivalent to training an ensemble of 2^n sub-networks with shared weights, and explain why shared weights make this tractable.
- (b) Using an expected value argument, derive what scaling must be applied at test time to ensure the expected output of each neuron matches its training behaviour. What observable symptom arises if this correction is not applied?
- (c) Given the transform used in batch normalization, $A' = \frac{A - \mu}{\sqrt{\sigma^2 + \epsilon}}$ followed by $\alpha A' + \beta$, explain why batch normalization has a regularizing effect and when you would prefer dropout over it.

Solution:

(a) With $n = 4$ neurons and $p = 0.5$, each neuron is independently either kept or dropped, giving $2^4 = 16$ distinct sub-networks. Training with dropout is equivalent to training an ensemble because at each mini-batch a different sub-network is sampled and updated, so over many iterations all 2^n sub-networks are implicitly trained. Shared weights make this tractable because all sub-networks share the same parameter vector θ without weight sharing, training 2^n independent networks would be computationally infeasible for large n .

(b) During training, each neuron is active with probability $(1 - p)$, so its expected output is $(1 - p) \cdot h$, where h is its full output. At test time the full network is used without dropout, so the expected output is h . To match training behaviour, the learned weights must be multiplied by $(1 - p)$ at test time, scaling the output down to $(1 - p) \cdot h$. If this correction is not applied, the test-time activations will be larger than what the network saw during training, causing overconfident predictions and degraded performance.

(c) Batch normalization normalizes each layer's activations to zero mean and unit variance via $A' = \frac{A - \mu}{\sigma}$, then rescales with learnable parameters α and β . This has a regularizing effect because normalizing across a mini-batch introduces noise into the activations (since μ and σ are computed per batch and vary across batches), which prevents neurons from overfitting to specific input patterns similar to dropout's effect. Dropout is preferred over batch normalization when the batch size is very small (making batch statistics unreliable).

6. [CO1, CO2, CO3] **Train-Test split, Cross Validation** (3 points)

A student is building an EEG-based epilepsy detector. The dataset has 20 subjects: 15 healthy and 5 epileptic. Each subject has 3 EEG recordings $\times$ 20 windows = 60 samples per subject, yielding 1200 samples in total.

- (a) The student splits the 1200 samples randomly into 80% train and 20% test. Explain whether the student's split is correct or not with explanation. If the student's approach is not right, then state the appropriate approach.
- (b) Suppose the student wants to perform k -fold cross-validation. Using the correct splitting approach from part (a) in a stratified manner, derive the number of samples used for training and testing in each fold, along with the class distribution of the test set, all expressed in terms of k .
- (c) Argue whether AUROC or AUPRC is the more appropriate evaluation metric in this setting, with justification.

Solution:

(a) The random split is incorrect because samples from the same subject are highly correlated. A random split places samples from the same subject in both train and test sets, causing data leakage. The correct approach is to split at the subject level to ensure that all samples from a subject are entirely in either train or test.

(b) Applying the correct subject-level split from part (a), there are 20 subjects in total. In each fold, $\frac{20}{k}$ subjects are held out for testing and $\frac{20(k-1)}{k}$ subjects are used for training. Since each subject contributes 60 samples, this yields $\frac{1200}{k}$ test samples and $\frac{1200(k-1)}{k}$ training samples per fold. To maintain stratification, each test fold contains $\frac{15}{k}$ healthy and $\frac{5}{k}$ epileptic subjects (so k must divide both 15 and 5, e.g. $k = 5$).

(c) AUPRC is more appropriate because the dataset is imbalanced (fewer epileptic subjects), and the task focuses on detecting the positive class. AUROC can be overly optimistic under class imbalance, whereas AUPRC better reflects performance on the minority (epileptic) class. Ideally, evaluation should also be performed at the subject level.

This course is offered in two tracks: **3XX (undergraduate)** and **5XX (graduate)**. Attempt **only one** of the following questions based on the course code written in the question.

Note: Attempting both questions or the question not corresponding to your course code **will not** fetch extra marks.

7. [CO1, CO2, CO4] **Support Vector Machines** [3XX] (3 points)

Consider the hard-margin SVM primal problem:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{x}_i^T \mathbf{w} + b) \geq 1 \quad \forall i = 1, 2, \dots, n$$

where $\mathbf{w} \in \mathbb{R}^p$ is the weight vector, b is the bias, $\mathbf{x}_i \in \mathbb{R}^p$ is the feature vector of the i -th sample, and $y_i \in \{-1, +1\}$ is its class label.

- (a) Since any rescaling $(\mathbf{w}, b) \rightarrow (\alpha \mathbf{w}, \alpha b)$ for $\alpha > 0$ preserves separability, explain why the unconstrained problem is ill-posed, and show that fixing $\min_i y_i(\mathbf{x}_i^T \mathbf{w} + b) = 1$ resolves this and leads to the equivalent formulation of maximizing the geometric margin $\frac{1}{\|\mathbf{w}\|}$.
- (b) Both logistic regression and hard-margin SVM produce a linear decision boundary of the form $\mathbf{x}_i^T \mathbf{w} + b = 0$ for a linearly separable dataset. A student argues that since both models yield the same form of decision boundary, their solutions $\mathbf{w}^*$ must be identical. Using your result from (b) and the nature of the respective loss functions, argue whether the student is correct or not.

Solution:

(a) Without any constraints, the problem has no unique solution because you can always scale $(\mathbf{w}, b)$ up or down and keep the data separated, which means the margin $\frac{1}{\|\mathbf{w}\|}$ can grow without bound.

To fix this, we pin the closest points to the boundary by requiring $\min_i y_i(\mathbf{x}_i^T \mathbf{w} + b) = 1$. This forces the support vectors to sit exactly on the lines $\mathbf{x}_i^T \mathbf{w} + b = \pm 1$.

Deriving the geometric margin. Take a support vector $\mathbf{x}_+$ on L_{+1} , so $\mathbf{x}_+^T \mathbf{w} + b = 1$. Its perpendicular distance to the decision boundary $L_0 : \mathbf{x}^T \mathbf{w} + b = 0$ is:

$$d_+ = \frac{|\mathbf{x}_+^T \mathbf{w} + b|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

By symmetry, a support vector $\mathbf{x}_-$ on L_{-1} satisfies $\mathbf{x}_-^T \mathbf{w} + b = -1$, which gives:

$$d_- = \frac{|\mathbf{x}_-^T \mathbf{w} + b|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

The total geometric margin between L_{+1} and L_{-1} is therefore:

$$\text{margin} = d_+ + d_- = \frac{2}{\|\mathbf{w}\|}$$

Maximizing $\frac{2}{\|\mathbf{w}\|}$ is equivalent to minimizing $\|\mathbf{w}\|$, or equivalently $\frac{1}{2}\|\mathbf{w}\|^2$, subject to all points being correctly classified under the canonical scaling:

$$y_i(\mathbf{x}_i^T \mathbf{w} + b) \geq 1 \quad \forall i$$

(b) The student is incorrect. Both models share the same boundary form but optimize different objectives, so their $\mathbf{w}^*$ generally differ. The SVM maximizes the geometric margin $\frac{2}{\|\mathbf{w}\|}$, making the solution sensitive only to the support vectors. All other points have no effect on $\mathbf{w}^*$.

Logistic regression minimizes $\sum_i \log(1 + e^{-y_i(\mathbf{x}_i^T \mathbf{w} + b)})$, which is never exactly zero for finite $\mathbf{w}$, so every point always contributes a non-zero gradient regardless of its distance from the boundary. The SVM boundary is determined by a few support vectors whereas logistic regression's $\mathbf{w}^*$ is optimized by the entire dataset therefore, the solutions will not be the same.

7. [CO1, CO2, CO4] Support Vector Machines [5XX]

(3 points)

The SVM loss function $\mathcal{L}'(\boldsymbol{\theta})$ is defined as

$$\mathcal{L}'(\boldsymbol{\theta}) = \eta \sum_{i=1}^n \left[y_i C_1(z_i) + (1 - y_i) C_0(z_i) \right] + \frac{1}{2} \sum_{j=1}^p \theta_j^2,$$

where $z_i = \mathbf{x}_i^T \boldsymbol{\theta} + b$, $C_1(z_i) = \max(0, 1 - z_i)$, and $C_0(z_i) = \max(0, 1 + z_i)$.

- (a) Show that when $\eta \rightarrow \infty$ and $a(\boldsymbol{\theta}) = 0$, the constraints reduce to $y_i(\mathbf{x}_i^T \boldsymbol{\theta} + b) \geq 1$ for all i , where $y_i \in \{-1, +1\}$. Hence state the hard-margin SVM primal and express the geometric margin in terms of $\|\boldsymbol{\theta}\|$.
- (b) Two support vectors $\mathbf{x}_1 = (1, 1)^T$, $y_1 = +1$ and $\mathbf{x}_2 = (-1, -1)^T$, $y_2 = -1$ lie on the margin boundaries with $b = 0$. Using the support vector constraints with equality, find $\boldsymbol{\theta}^*$, compute the geometric margin, and verify that minimising $\frac{1}{2}\|\boldsymbol{\theta}\|^2$ is equivalent to maximising this margin.

Solution: (a) When $\eta \rightarrow \infty$, any non-zero hinge loss incurs an unbounded penalty, so at optimality both C_1 and C_0 must equal zero.

- $y_i = +1$: $\max(0, 1 - z_i) = 0 \Rightarrow z_i \geq 1$
- $y_i = -1$: $\max(0, 1 + z_i) = 0 \Rightarrow z_i \leq -1$

Multiplying the second equation by $y_i = -1$ unifies both cases to $y_i(\mathbf{x}_i^\top \boldsymbol{\theta} + b) \geq 1$. The hard-margin primal is:

$$\min_{\boldsymbol{\theta}, b} \frac{1}{2} \|\boldsymbol{\theta}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{x}_i^\top \boldsymbol{\theta} + b) \geq 1 \quad \forall i.$$

A support vector on L_{+1} has distance $\frac{1}{\|\boldsymbol{\theta}\|}$ to L_0 , and by symmetry so does one on L_{-1} , giving a total geometric margin of $\frac{2}{\|\boldsymbol{\theta}\|}$. Minimising $\frac{1}{2} \|\boldsymbol{\theta}\|^2$ directly maximises this.

(b) Applying the support vector constraints with equality and $b = 0$:

For $\mathbf{x}_1 = (1, 1)^\top$, $y_1 = +1$:

$$y_1(\mathbf{x}_1^\top \boldsymbol{\theta}) = (+1) \cdot (1, 1) \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \theta_1 + \theta_2 = 1.$$

For $\mathbf{x}_2 = (-1, -1)^\top$, $y_2 = -1$:

$$y_2(\mathbf{x}_2^\top \boldsymbol{\theta}) = (-1) \cdot (-1, -1) \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \theta_1 + \theta_2 = 1.$$

Both constraints are identical, so we minimise $\frac{1}{2}(\theta_1^2 + \theta_2^2)$ subject to $\theta_1 + \theta_2 = 1$. By symmetry:

$$\boldsymbol{\theta}^* = \left(\frac{1}{2}, \frac{1}{2}\right)^\top, \quad \|\boldsymbol{\theta}^*\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{\sqrt{2}}.$$

The geometric margin is:

$$\frac{2}{\|\boldsymbol{\theta}^*\|} = \frac{2}{1/\sqrt{2}} = 2\sqrt{2}.$$

Since $\boldsymbol{\theta}^*$ minimises $\frac{1}{2} \|\boldsymbol{\theta}\|^2 = \frac{1}{4}$, it simultaneously maximises the margin $2\sqrt{2}$, which confirms the equivalence.
